Signed order count in aggregate price impact

TSLA, 2019

Order reconstruction, chronological validation, and scaling replication

Riad Darwish

CentraleSupélec

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This analysis began in the course project *Liquidity Games in High-Frequency Markets*, supervised by Anastasia Bugaenko of Capital Fund Management. The course presentation was prepared by L. Cohen, R. Darwish, A. Pariente, H. Rohrbach, and R. Sithisak. I later rewrote the supervised analysis and prepared this report in a separate repository.

Abstract

I reconstruct 4,573,957 visible market orders from 252 TSLA sessions in 2019. LOBSTER messages are aligned with the book state immediately before each event, and fills with the same timestamp and initiating side are grouped. The main experiment relates signed volume and signed order count in a 10-order window to the mid-price change over that window. Since the regressors are observed during the window, the exercise estimates conditional impact rather than a pre-trade forecast.

The final 20 percent of dates form an untrimmed holdout of 102,495 windows. Signed-volume OLS reaches $R^2 = 0.118$. Nonlinear volume terms raise the score to 0.298, and raw signed order count raises it to 0.359. Further count transforms change little: the full score is 0.360. Mean squared error falls by 27.4% relative to the linear baseline. Resampling complete test dates gives a 95 percent interval of 26.5% to 28.4% for that reduction.

I also apply the finite-size scaling construction of Patzelt and Bouchaud (2018) to the same stock-year. The width and height scale regressions have in-sample R^2 values of 0.9963 and 0.9978. These fits use 28 estimated scale points and are unrelated to the chronological prediction score.

Keywords: market impact, order-flow imbalance, market microstructure, chronological validation, finite-size scaling

1 Question

In the local benchmark of Kyle (1985), signed order flow and price change are connected by a linear slope. Empirical impact curves bend, change scale with the aggregation horizon, and reflect persistent trade signs (Bouchaud et al., 2018; Patzelt and Bouchaud, 2018). I focus on one question: for TSLA in 2019, does signed order count explain price impact on later dates after signed volume has already been included?

Signed volume alone explains 0.118 of the holdout variance. Fixed volume transforms explain 0.298. Adding raw signed count brings the score to 0.359, almost the same as the full specification at 0.360. Raw signed count provides the gain. Extra transforms and ridge regression leave the reported score unchanged.

Both regressors and response cover the same event-time window. The date split tests whether a conditional mapping estimated earlier in the year transfers to later dates. It does not identify a causal effect because order flow and liquidity react to each other.

The primary holdout keeps every observation. I estimate uncertainty by resampling trading dates, then repeat the comparison over expanding test blocks, five window lengths, and a basis-point response. The earlier trimmed protocol is reported separately because it produced the original 0.24 to 0.38 comparison.

2 Relation to prior work

Kyle's model gives a local liquidity measure: a larger price response per unit of signed flow corresponds to a less deep market (Kyle, 1985). It does not imply that the empirical response must remain linear far from balanced flow. The evidence reviewed by Bouchaud et al. (2018) shows why impact is difficult to see at the trade level and why conditional averaging is central to measurement.

Patzelt and Bouchaud (2018) study volume and sign imbalance over several intraday horizons. Their conditional curves can be rescaled by a horizon-dependent width and height. The published claim is based on many instruments and periods. Here, the same functional form is fitted to TSLA in 2019 only. The replication can say whether that form describes this stock-year. It cannot reproduce the paper's cross-sectional evidence.

The course project followed Bugaenko (2019), which connects order-flow imbalance, local liquidity estimates, and supervised impact models. I retain the transaction reconstruction and scaling exercise, but rebuild the supervised evaluation around complete chronological samples. The model ablation isolates signed count as the useful extension of the volume baseline.

3 Data and transaction reconstruction

3.1 LOBSTER alignment

LOBSTER provides a message file and an order-book file for each session. Message row k is the event that moves the book from row $k-1$ to row k (LOBSTER, 2026). The pre-event best bid and ask for an execution on row k therefore come from order-book row $k-1$. The pipeline makes that shift before calculating the mid-price or spread. An execution on the first row of a session is discarded because its pre-event state is absent.

Prices in the source files are integers scaled by 10,000. For a visible execution, event type 4, LOBSTER's direction field identifies the resting order. The aggressor sign is its opposite: $+1$ for a buyer-initiated execution and -1 for a seller-initiated execution.

A market order may consume several displayed orders. The source used here does not expose a parent market-order identifier, so visible fills with the same date, timestamp, and aggressor sign are grouped. Share size is summed, execution price is volume weighted, and the first pre-event book defines the starting state. This is a reconstruction rule, not investor identification.

3.2 Sample

Table 1 reports descriptive quantities, not model results. The median reconstructed order has 54 shares. The distribution is right-skewed: its 90th and 99th percentiles are 194 and 700 shares.

The median quoted spread is 7.0 cents, and the share of buyer-initiated orders is 49.7%.

Table 1. Reconstructed visible-order sample. Daily quantiles are computed across 252 sessions; order-level quantiles use all reconstructed visible market orders.

Quantity	Value
Timestamp-aggregated visible market orders	4,573,957
Median orders per session	16,034
10th to 90th percentile, orders per session	10,752 to 29,642
Median order size, shares	54
90th and 99th percentile order size, shares	194, 700
Median and 90th percentile spread, cents	7.0, 15.0
Buyer-initiated fraction	49.7%
Non-overlapping 10-order windows	457,263
Standard deviation of 10-order impact, cents	12.47
Zero-impact share of 10-order windows	3.0%

The raw and derived row-level files remain outside the repository because the LOBSTER license does not permit redistribution. Committed CSV and JSON files contain aggregate curves, scores, and sample summaries only.

3.3 A separate scaling table

The supervised experiment uses visible executions because their aggressor side is given by the resting-order direction. The scaling exercise also uses hidden executions, event type 5, to match the broader transaction definition in Patzelt and Bouchaud (2018). Their signs are inferred from execution price relative to the pre-event mid-price. Midpoint executions have no inferred side and are removed. After timestamp grouping, this table contains 5,879,277 transactions. The prediction and scaling exercises therefore use different transaction tables.

4 Measurement

Consider a same-session window beginning at reconstructed market order t and containing T signed orders. With aggressor sign ϵ_i, share size v_i, and pre-event mid-price m_i, define

$$\Delta V_{t,T} = \sum_{i=t}^{t+T-1} \epsilon_i v_i, \tag{1}$$

$$\Delta E_{t,T} = \sum_{i=t}^{t+T-1} \epsilon_i, \tag{2}$$

$$Y_{t,T} = 100\,(m_{t+T} - m_t)\,. \tag{3}$$

Here, Y is measured in cents. Windows are non-overlapping within each horizon and never cross a session boundary. A T-order window needs the mid-price at its right edge, so the final incomplete block of each session is discarded.

The count imbalance ΔE distinguishes flow patterns that signed volume alone merges. One large buy and several smaller buys can generate comparable ΔV but different ΔE. That difference may reflect order splitting, replenishment, or state-dependent order size. The regression cannot separate those mechanisms, but it can test whether the count adds conditional information on dates not used for estimation.

Figure 1 displays the raw conditional mean at four horizons. Quantile bins are used only for the figure. The outer bins contain wide ranges of volume, while the central bins reveal a steep response around balanced flow. The flattening in both tails is visible at every horizon. A single global linear coefficient cannot capture this shape.

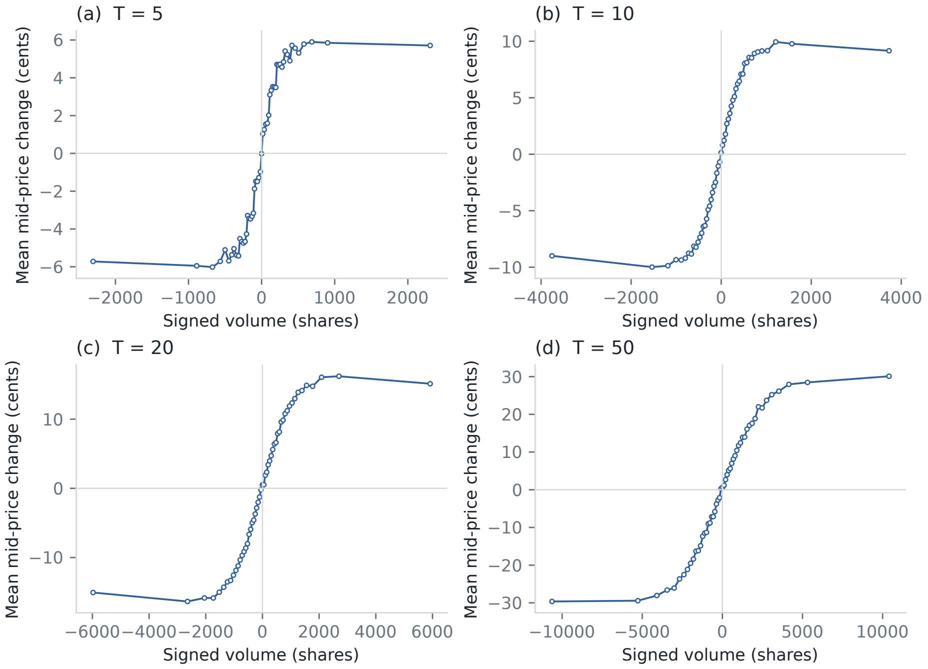

Figure 1. Mean mid-price change conditional on signed volume. Each point is a mean within one of 51 signed-volume quantiles. All 252 dates enter; the bins are not used for model fitting.

A local Kyle slope is also estimated within the central 35 percent of absolute signed volume. At $T = 10$, the fitted value is 0.01837 cents per share. Across $T \in \{5, 10, 20, 50, 100\}$, the fitted slopes decay approximately as $T^{-0.313}$. This is a descriptive liquidity estimate, not a trade-level execution-cost model.

5 Evaluation design

5.1 Chronological holdout

The main $T = 10$ sample contains 457,263 windows across 252 dates. The first 80 percent of distinct dates form the training partition; the remaining dates are held out. Table 2 gives the exact boundary. No row is removed from either side of the split.

Table 2. Primary chronological partition.

Partition	First date	Last date	Windows
Train	2019-01-02	2019-10-17	354,768
Holdout	2019-10-18	2019-12-31	102,495

The four nested OLS specifications are:

$$\begin{aligned}
&\mathcal{M}_1 : \Delta V, \\
&\mathcal{M}_2 : \Delta V, |\Delta V|, \operatorname{sgn}(\Delta V)\sqrt{|\Delta V|}, \operatorname{sgn}(\Delta V)\log(1+|\Delta V|), \\
&\mathcal{M}_3 : \mathcal{M}_2, \Delta E, \\
&\mathcal{M}_4 : \mathcal{M}_3, |\Delta E|, \operatorname{sgn}(\Delta E)\sqrt{|\Delta E|}.
\end{aligned}$$

The transforms are fixed before estimation. An intercept is included. Ridge regression with standardized $\mathcal{M}_4$ features is retained as a numerical check, but its predictions are indistinguishable from OLS at the shown precision.

Scores are calculated only on the holdout:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2, \tag{4}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{5}$$

Mean absolute error and sign accuracy are reported as secondary diagnostics. Sign accuracy omits a window when either the realized or fitted response is exactly zero.

5.2 Uncertainty and stability checks

Event-time windows from one session share market conditions. Treating all windows as independent would understate uncertainty. The bootstrap therefore resamples the 51 complete holdout dates with replacement. For each of 10,000 repetitions, scores are recomputed from the selected daily blocks. The interval is the 2.5th to 97.5th percentile of the bootstrap distribution.

Five additional temporal splits begin with 40 percent of the dates for training. Each subsequent block is tested once, then added to the expanding training set. Horizon checks repeat the 80/20 split for $T = 5, 10, 20, 50, 100$. Finally, the response is replaced by $10{,}000 \log(m_{t+T}/m_t)$ to check that the result is not an artifact of measuring a changing stock price in cents.

6 Holdout results

6.1 Feature ablation

Table 3 separates curvature from count information. The linear volume baseline has $R^2 = 0.118$. Fixed volume transforms account for the first increase, to 0.298. Adding only raw signed count reaches 0.359. The two additional count transforms change the third decimal place by less than one point. Raw signed count accounts for the useful part of the second increase.

Table 3. Full chronological holdout, $T = 10$. Errors are measured in cents.

Specification	R^2	MSE	MAE	Sign accuracy
Signed volume	0.118	147.23	8.64	77.0%
Volume transforms	0.298	117.18	7.79	76.9%
Volume transforms plus raw signed count	0.359	106.96	7.48	78.5%
Volume and count transforms	0.360	106.89	7.47	78.5%

Figure 2 compares fitted and realized means in 20 holdout bins. The linear volume model spans too narrow a range for the S-shaped response. The count model follows the binned means more closely, although individual windows remain noisy.

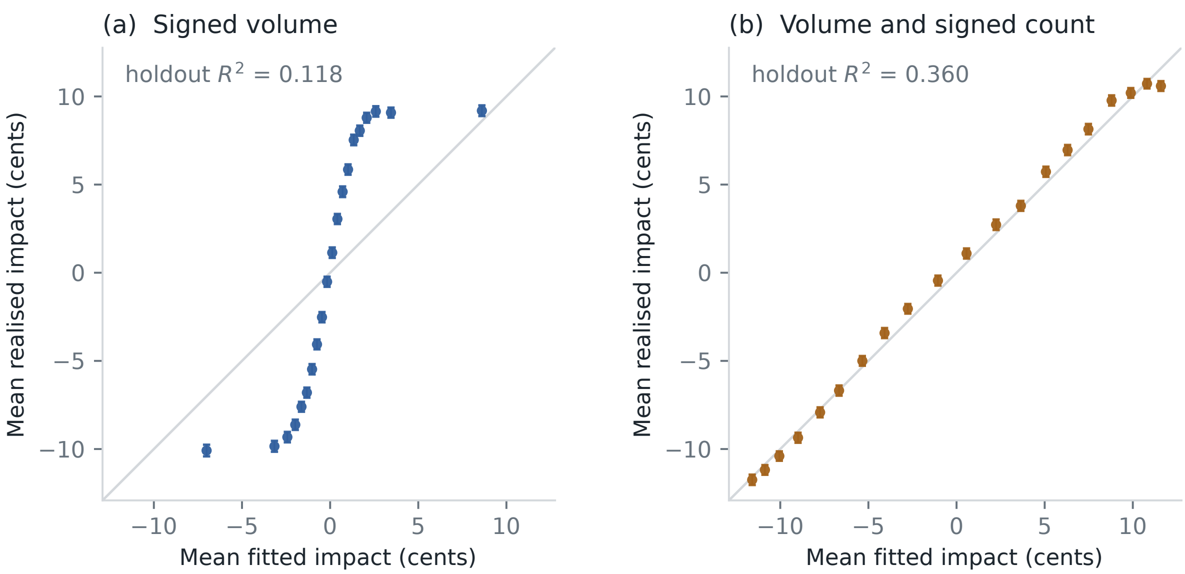

Figure 2. Holdout calibration by fitted-value quantile. Points show mean fitted and realized impact in 20 equally populated bins. Error bars are 1.96 observation-level standard errors and are included only as a visual guide; the formal score interval is clustered by date.

Figure 3 groups the holdout residual of the volume model by signed count. The mean moves from roughly −4 cents for ten sells to more than 3 cents for strong buy counts. This monotone residual pattern is the signal captured by ΔE.

Figure 3. Holdout residual from the volume-transform model, grouped by signed order count. Points are equal-weighted means across dates. The band is 1.96 standard errors of the date-level means.

The augmented model lowers MSE from 147.23 to 106.89 cents squared, a relative reduction of 27.4%. The day-clustered 95 percent interval is [26.5%, 28.4%]. The corresponding increase in R^2 is 0.242, with interval [0.232, 0.252]. None of the 10,000 bootstrap draws gives a non-positive improvement. This interval quantifies sampling variation across the 51 late-year dates. It does not cover model-selection or cross-year uncertainty.

6.2 Time and horizon checks

The count specification outperforms signed volume in every expanding test block in Figure 4. Its R^2 ranges from about 0.33 to 0.38, while the baseline ranges from about 0.10 to 0.14. The gap is present before the final December block.

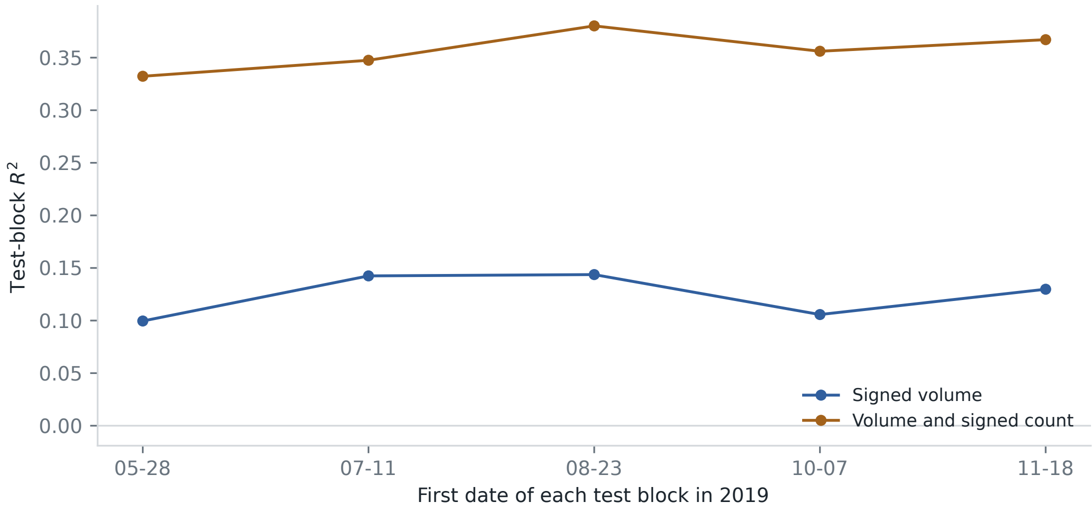

Figure 4. Five expanding-window evaluations. Each point is scored on the contiguous block beginning at the date shown. Earlier test blocks are added to the training set before the next fit.

The same ordering holds from 5 to 100 orders (Figure 5). At $T = 5$, R^2 rises from 0.074 to 0.282. At $T = 100$, it rises from 0.269 to 0.482. Longer windows average more idiosyncratic trade-level noise and contain a wider count range, so higher scores at larger T should not be read as better short-horizon forecasts.

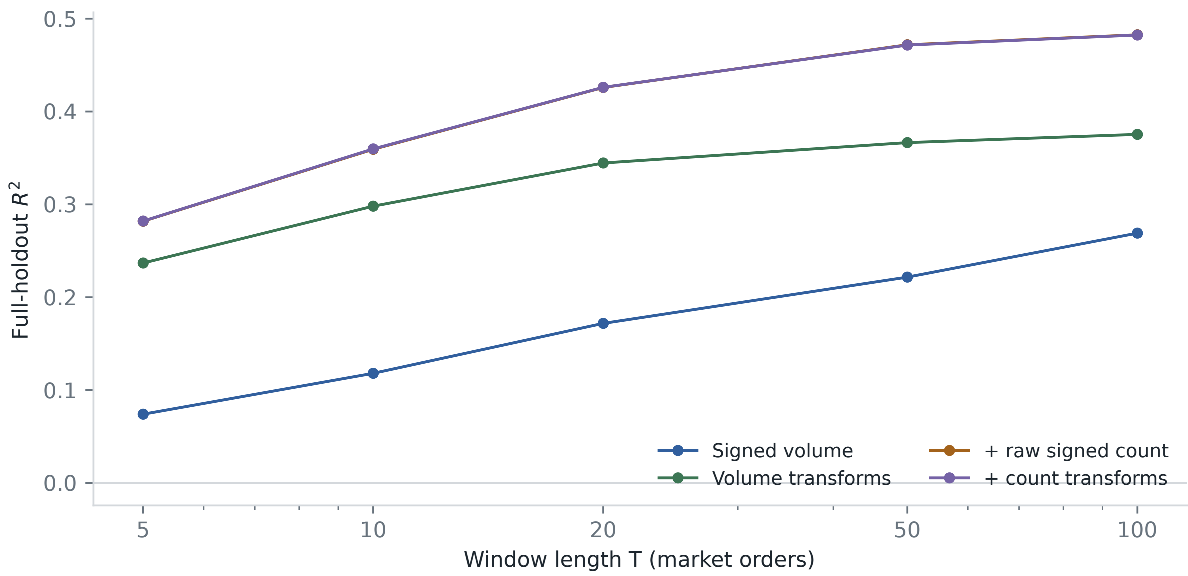

Figure 5. Full-holdout R^2 across event-time horizons. The raw count term accounts for nearly all of the gap between the volume-transform and full specifications.

The holdout result is also present within each calendar month. The augmented R^2 is 0.329 in the October portion, 0.378 in November, and 0.363 in December. With log return in basis points as the response, the scores are 0.114 and 0.309, and MSE falls by 22.0%. Price normalization reduces the size of the gain but does not remove it.

6.3 Why the previous score was 0.24 to 0.38

The earlier course protocol removed observations beyond three standard deviations in both signed volume and realized impact. Re-estimating those bounds on training dates avoids using the test distribution to set the cutoff, but applying the impact cutoff to the test still selects observations by the realized response. Under that trimmed comparison, the baseline has $R^2 = 0.241$ and the augmented model has $R^2 = 0.383$. MSE falls by 18.7%.

The trimmed scores describe the retained central sample. Selection on realized test impact makes them unsuitable as the primary evaluation. The cutoff raises the baseline much more than the count model, explaining the difference between 0.24 to 0.38 and the untrimmed 0.12 to 0.36 comparison.

7 Scaling replication

7.1 Construction

The scaling exercise follows the aggregate-volume form in Patzelt and Bouchaud (2018). For a window of N timestamp-aggregated type-4 and type-5 transactions, let Q be signed share volume after a daily-volume adjustment, and let r be the log mid-price change. Conditional curves are estimated in 31 quantile bins for 28 log-spaced horizons from 10 to 5,000 transactions.

Each curve is modeled as

$$\mathcal{R}_N(Q) = R_N F(Q/Q_N), \tag{6}$$

$$F(x) = \frac{x}{(1+|x|^\alpha)^{\beta/\alpha}}. \tag{7}$$

One width Q_N and height R_N are estimated per horizon, with a common α and β. The nonlinear fit uses observation-count weights and a soft-L1 loss. Figure 6 plots the resulting rescaled curves. The collapse is close over the central observed range, with more dispersion around the origin and at a few long-horizon curves.

7.2 Scale fits and interpretation

After the curve fit, the scale parameters are regressed on horizon in log space:

$$Q_N = a_Q N^\xi, \qquad R_N = a_R N^\psi. \tag{8}$$

Table 4 and Figure 7 report those second-stage fits. The width exponent is 0.976 and the height exponent is 0.656. The residual panels show that the largest horizons carry the main deviations from a straight power law.

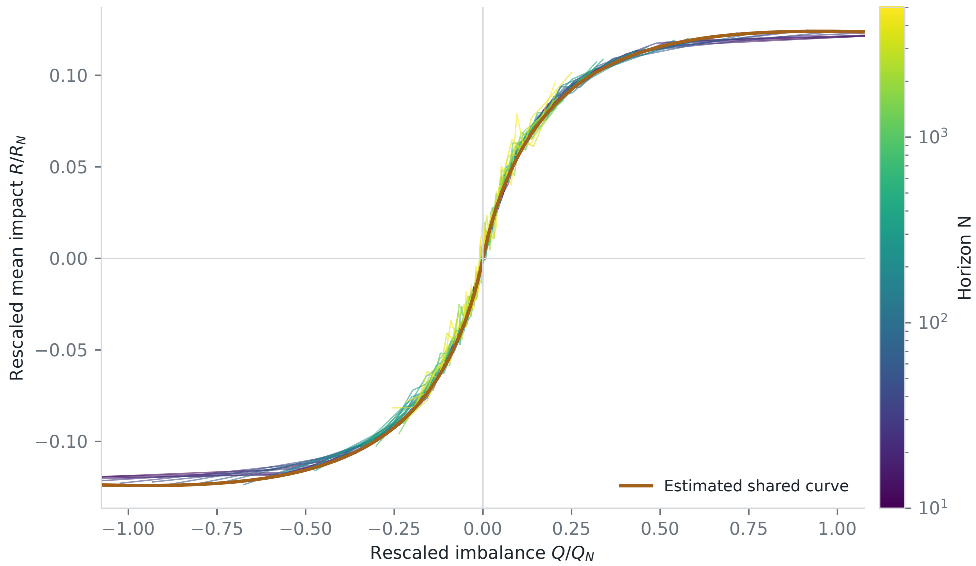

Figure 6. Finite-size scaling collapse for TSLA 2019. Color identifies the aggregation horizon. The orange line is the common fitted shape. This is an in-sample curve construction, not a forecast evaluation.

Table 4. Log-log fits to the 28 estimated scale parameters. Standard errors condition on the first-stage scale estimates.

Scale	Exponent	Standard error	In-sample R^2
Width Q_N	0.976	0.012	0.9963
Height R_N	0.656	0.006	0.9978

Near zero imbalance, the master-curve slope scales as R_N/Q_N. The fitted exponents imply a decay $N^{-(\xi-\psi)}$ with $\xi - \psi = 0.320$. The local Kyle slopes in Section 4 decay with exponent 0.313. The two calculations use different transaction samples and return definitions, so their agreement is descriptive.

The order-sign standard deviation gives a Hurst exponent of 0.693, in line with persistent trade signs in this sample. The 0.9963 and 0.9978 values belong to regressions on 28 estimated scale points. They do not measure future price variation. This dataset supports a replication of the proposed scaling form on TSLA 2019; the cross-asset claim in Patzelt and Bouchaud (2018) requires a broader panel.

8 Limitations

- The variables and response cover the same window, and order flow is endogenous. The fitted coefficients are neither pre-trade forecasts nor causal price effects.
- Timestamp and side grouping cannot recover investor identity or institutional metaorders. The book covers NASDAQ rather than consolidated US liquidity, and hidden-execution signs are uncertain around the midpoint.
- The feature family grew out of the course analysis. Although dates are held out chronologically, there is no second untouched year after the feature comparison. The bootstrap does not account for that research process.

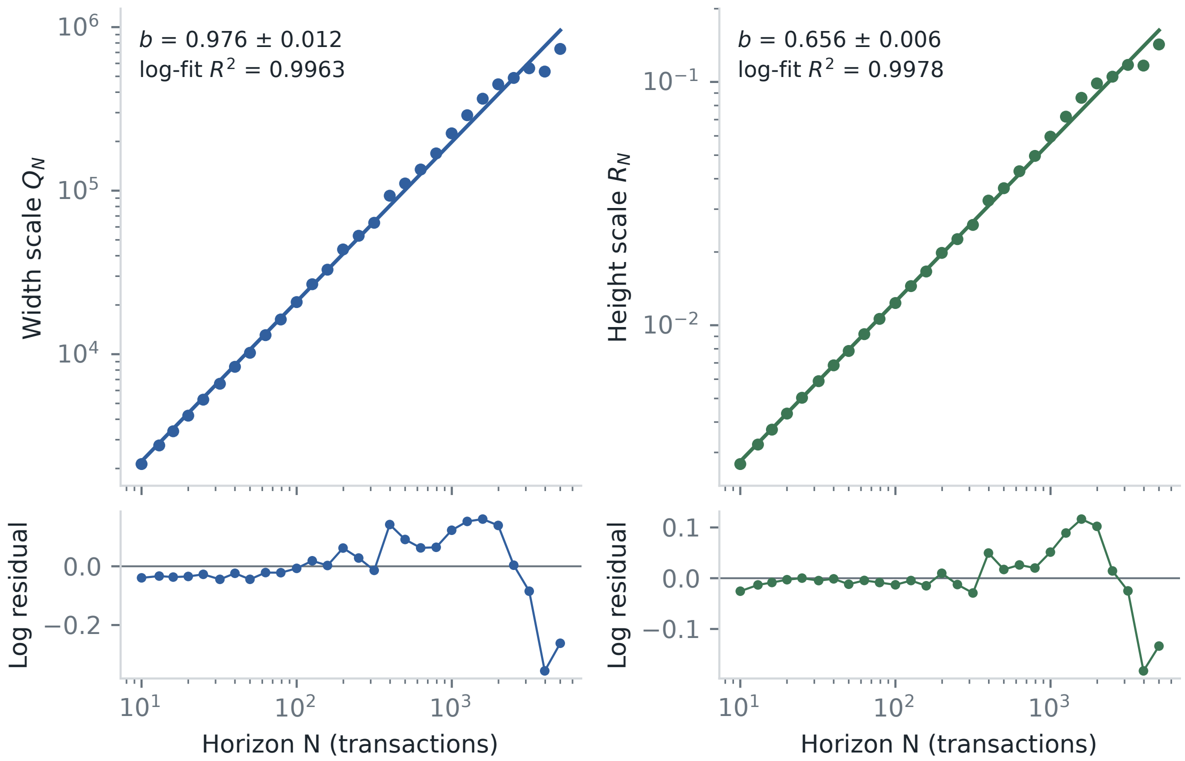

Figure 7. Power-law fits for the estimated width and height scales. Lower panels show log residuals. The displayed R^2 values describe these 28 in-sample points only.

- The scaling standard errors treat first-stage Q_N and R_N estimates as data. They do not propagate all curve-fitting uncertainty, and overlapping information across horizons remains.
- TSLA in 2019 is one liquidity regime. Neither the count result nor the fitted exponents can be assumed to hold for another asset or year.

9 Reproducibility and provenance

The Python package reconstructs the two transaction tables, builds event-time windows, fits the models, and writes the aggregate files under `results/`. The figures are vector PDFs. Synthetic tests cover file pairing, book alignment, day boundaries, chronological splits, clustered resampling, and the scaling regressions. Reproducing the annual tables requires licensed LOBSTER files, which are not tracked.

The course presentation lists L. Cohen, R. Darwish, A. Pariente, H. Rohrbach, and R. Sithisak. Anastasia Bugaenko supervised the work. I wrote this standalone package and report after the group project; the collaborative repository remains separate.

10 Conclusion

On the complete late-2019 holdout, adding signed count to the volume model moves R^2 from 0.118 to 0.360 and reduces MSE by 27.4%. Raw count accounts for almost all of that gain. The ordering is stable across the time and horizon checks reported above.

TSLA's 2019 conditional curves also fit the scaling form proposed by Patzelt and Bouchaud

(2018). That is a replication on one stock-year. The supervised model is evaluated by the chronological score.

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